1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

- **Devices using SMPS (Switch Mode Power Supply)**

a. Laptop Charger

- Uses SMPS because it efficiently converts AC mains power to the low-voltage DC required by the laptop with minimal heat loss.

b. Desktop Computer Power Supply

- Uses SMPS to provide multiple stable DC voltages (+12V, +5V, +3.3V) needed by computer components.

c. LED Television

- Uses SMPS because it delivers regulated power efficiently while keeping the TV compact and lightweight.

- **Devices Using UPS (Uninterruptible Power Supply)**

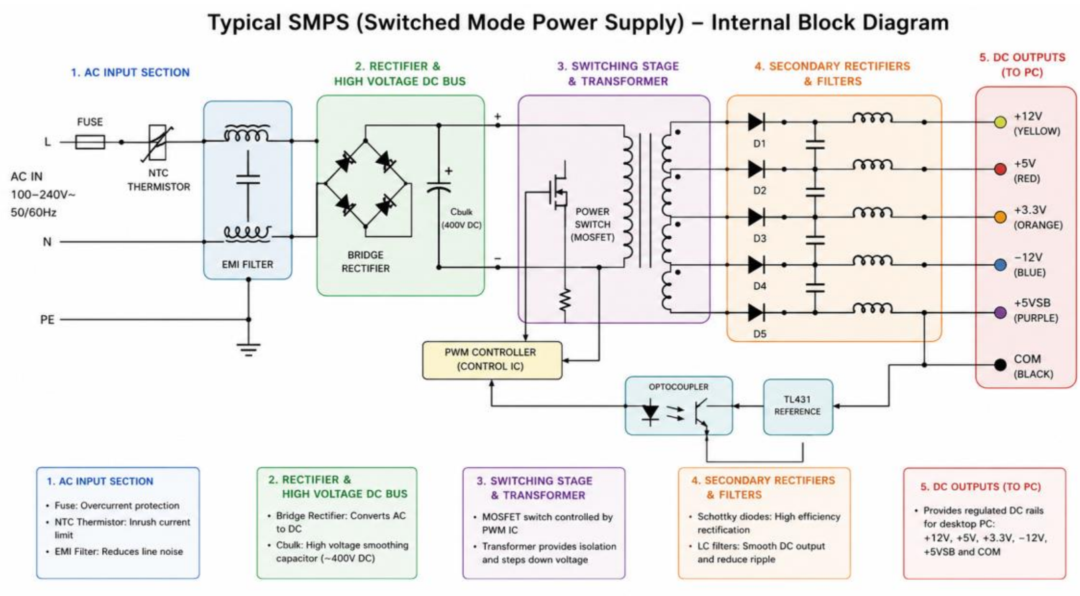
I. Desktop Computer with UPS Backup

- Uses a UPS to prevent sudden shutdowns and data loss during power outages.

II. Wi-Fi Router Connected to a UPS

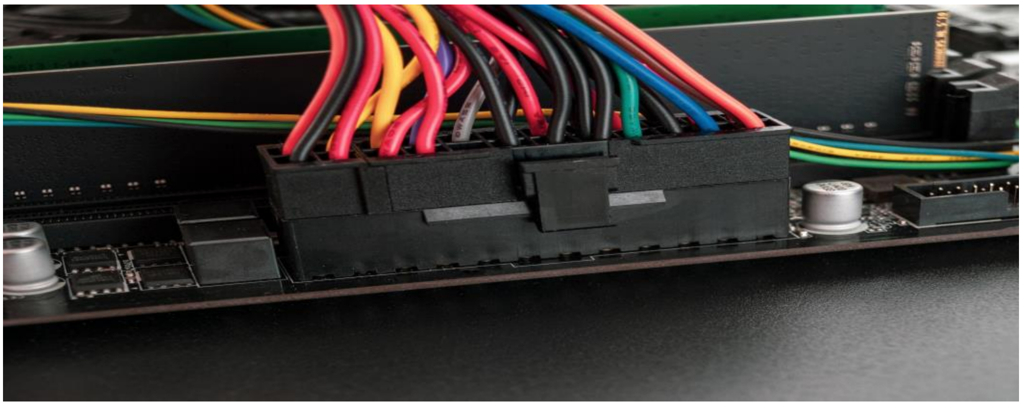
- Uses a UPS to maintain internet connectivity when the main power supply fails.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1) ATX 24-Pin Motherboard Connector



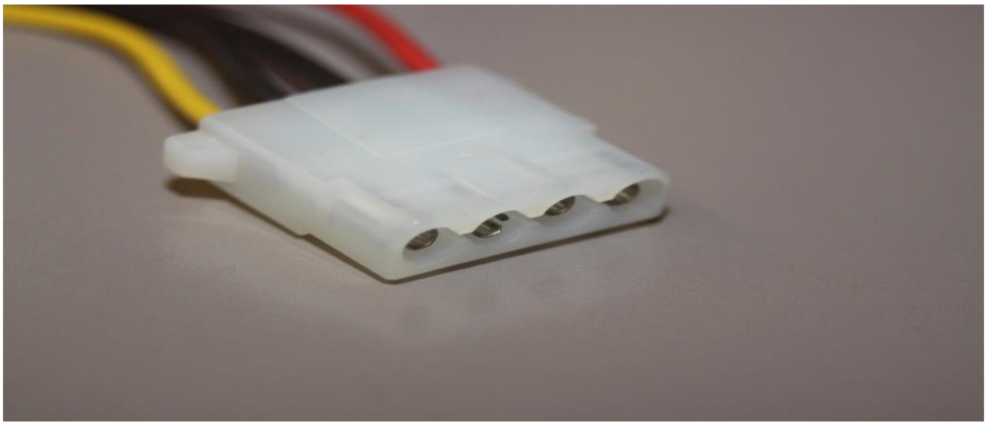
- **What it Powers**
- The main motherboard power connection. It supplies power to the motherboard, RAM slots, chipset, USB controllers, and other on-board components.
- **Safety Tip**
- Always shut down the PC and switch off/unplug the power supply before connecting or disconnecting it. Press the locking clip before removing the connector to avoid damaging the motherboard socket.

2) SATA Power Connector

- **What it Powers**
- SATA SSDs, SATA hard drives (HDDs), optical drives (DVD/Blu-ray drives), and some accessories such as fan hubs and RGB controllers.

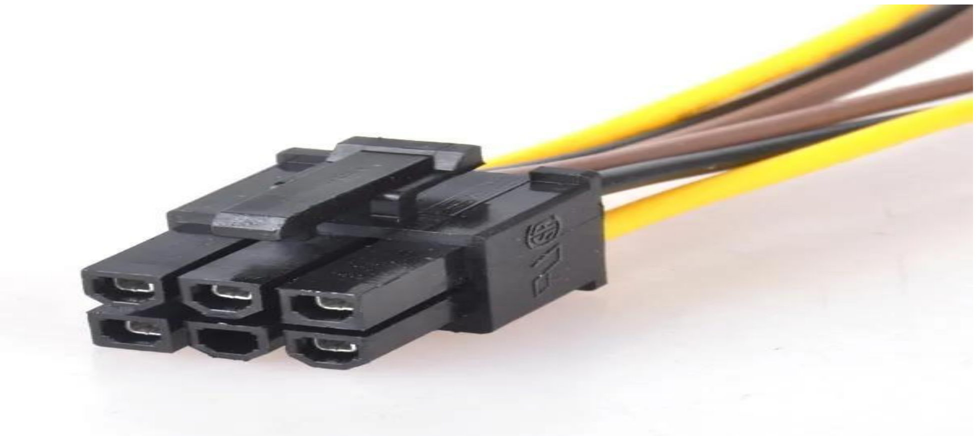
- **Safety Tip**
- Align the L-shaped key before inserting. Never force it; if it does not slide in easily, check the orientation.

3) **Molex 4-Pin Connector**



- **What it Powers**
- Older IDE/PATA hard drives, optical drives, case fans, fan controllers, and some legacy accessories.
- **Safety Tip**
- Pull on the plastic connector body, not the wires, when disconnecting it. Tugging the wires can loosen or damage the connections.

4) **PCle 6-Pin/8-Pin Connector**



- **What it Powers**
- Dedicated graphics cards (GPUs) and some high-performance expansion cards that require additional power beyond what the motherboard slot provides.

- **Safety Tip**
- Make sure the connector clicks into place and is fully seated before powering on the PC. A loose PCIe power connection can cause instability, overheating, or failure to boot.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

1. **Connect Power to the Motherboard**

- **Main Motherboard Power**
- Use the 24-pin ATX power cable from the PSU.
- Plug the 24-pin connector into the motherboard's main power socket.

Connector: 24-pin ATX

- **CPU Power**
- Use the 8-pin EPS/CPU power cable.
- Connect it to the CPU power socket near the CPU socket, usually at the top-left of the motherboard.
- Some high-end motherboards may require an additional 4-pin or 8-pin CPU power connector.

Connector: 8 pin EPS (4+4) CPU power

2. **Connect Power to the Graphic Card (GPU)**

- Install the graphics card into the PCIe x16 slot on the motherboard.
- Connect the appropriate PCIe power cables from the PSU to the GPU.

Common GPU Power Connectors:

- 6-pin PCle
- 8-pin PCle(6+2)
- Dual 8-pin PCle
- 12VHPWR or 12V-2x6 connector

Connector: PCIe power cable (6-pin, 8-pin, 6+2, or 12VHPWR depending on GPU)

3. Connect Power to SATA Storage Devices

For SATA SSDs and HDDs:

- Connect a SATA power cable from the PSU to each drive.
- Connect a separate SATA data cable from the drive to a SATA port on the motherboard.

Power Connector: SATA power (15-pin)

Data connector: SATA Data Cable